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Your Roll No.....

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Sr. No. of Question Paper : 5733

Unique Paper Code : 2352012301

Name of the Paper : Group Theory

Name of the Course : B.Sc. (H) Mathematics

Semester : III

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt all questions by selecting two parts from each question.
3. All questions carry equal marks.
4. Use of Calculator is not allowed.

1. (a) If a permutation $\sigma = \alpha\beta$ where α and β are disjoint cycles of length m and n respectively, then prove that order of σ equals $\text{lcm}(m, n)$.

Hence, find the order of the permutation $\sigma = (1235)(24567)$.

(b) Prove that the set of even permutations in S_n forms a subgroup of S_n . Also, show that $\alpha^{-1}\beta^{-1}\alpha\beta$ is an even permutation for $\alpha, \beta \in S_n$.

(c) Let H be a subgroup of G and $a \in H$. Prove that aH is a subgroup of G if and only if $a \in H$.

Also, find all the left cosets of $H = \{(1), (13)\}$ in S_3 .

2. (a) Let G be a finite group and H be a subgroup of G . Prove that $|H|$ divides $|G|$. Hence, prove that a group of prime order is cyclic.

(b) Prove that a subgroup H of G is normal in G if and only if $xHx^{-1} \subseteq H$, for all $x \in G$. Hence, show that the subgroup $SL(2, \mathbb{R})$ of 2×2 matrices with determinant 1 is a normal subgroup of $GL(2, \mathbb{R})$, the group of 2×2 matrices with non-zero determinant.

(c) Prove that $HK = \{hk \mid h \in H, k \in K\}$ is a subgroup of a group G if H is a normal subgroup of G and K be any subgroup of G . Does the result hold if H is not a normal subgroup? Justify.

P.T.O.

3. (a) State and prove First Isomorphism Theorem. Show that $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to $\mathbb{Z}_n$, for all $n \in \mathbb{N}$.
- (b) Determine all the homomorphisms from $\mathbb{Z}_{20}$ onto $\mathbb{Z}_8$. How many are there to $\mathbb{Z}_8$? Justify.
- (c) (i) Let N be a normal subgroup of a finite group G . Prove that order of the group element gN in G/N divides the order of g .
- (ii) Find $\text{Aut}(\mathbb{Z}_8)$.
4. (a) State and Prove Cayley's Theorem.
- (b) If Φ is an isomorphism from a group G onto a group G' . Then prove that
- Φ^{-1} is an isomorphism from group G' onto G .
 - Order of a is equal to order of $\Phi(a)$ for all a in G .
- (c) (i) Let G be a group. Prove that the mapping $\Phi(g) = g^{-1}$ for all g in G is an automorphism iff G is Abelian.
- (ii) Show that $U(8)$ is not isomorphic to $U(10)$.
5. (a) Determine the number of elements of order 10 and the number of cyclic subgroups of order 10 in the group $\mathbb{Z}_{100} \oplus \mathbb{Z}_{25}$.
- (b) Let G and H be finite cyclic groups. Prove that $G \oplus H$ is cyclic if and only if $|G|$ and $|H|$ are relatively prime.
- (c) Consider the groups $G = \{3^a 6^b 10^c \mid a, b, c \in \mathbb{Z}\}$ under multiplication and $H = \{3^a 6^b 12^c \mid a, b, c \in \mathbb{Z}\}$ under multiplication. Prove that $G = \langle 3 \rangle \times \langle 6 \rangle \times \langle 10 \rangle$, whereas $H \neq \langle 3 \rangle \times \langle 6 \rangle \times \langle 12 \rangle$.
6. (a) (i) What is the smallest positive integer n such that there are two non-isomorphic groups of order n ? Name the two groups.
- (ii) Suppose that the order of some finite Abelian group is divisible by 10. Prove that the group has a cyclic subgroup of order 10.
- (b) The set $\{1, 9, 16, 22, 29, 53, 74, 79, 81\}$ is a group under multiplication modulo 91. Determine the isomorphism class of this group.
- (c) Prove, by comparing orders of elements, that the group $\mathbb{Z}_8 \oplus \mathbb{Z}_8$ is not isomorphic to the group $\mathbb{Z}_4 \oplus \mathbb{Z}_4$.